Sardar Patel Institute of Technology

Bhavan's Campus, Munshi Nagar, Andheri (W), Mumbai: 400058, India
(Autonomous College of Affiliated to University of Mumbai)

Special Examination
2023-24

Maxi Marks: 100 Duration: 3 hours
Class: T.E Semester: V
Course code: CS301 Branch: COMP/CSE-AIML/CSE-DS
Name of the course: Theory of Computation

Instructions:
(1) All questions are compulsory
(2) Partial credit will be given, so show your work where appropriate.
(3) Note: DFA = Deterministic Finite Automata, NFA = Nondeterministic Finite Automata, PDA = Push-Down Automata, CFG=Context Free Grammar, TM = Turing Machine.

Q No		Max Marks	CO	BL
Q1 A	Design a finite-state machine that will output the remainder when you divide a binary number by 4. Validate the machine based on the user's input.	10	1	3
B	Design a Moore machine to generate the output A if string ending with abb, B if string ending with aba, and C otherwise over alphabet (a, b). and convert it into Mealy machine. **OR** Construct a NFA for string ending with abb over the alphabet (a, b) and convert it into DFA.	10	1	3
Q2 A	Using pumping lemma proves that the following language is not regular. 1. $L = \{0^n 1^{n+1} \mid n > 0\}$ 2. A palindrome is a string that equals its own reverse, such as 0110 or 1011101. Prove that set of palindromes is not a regular language.	10	2	3
B	Convert the following grammars to Greibach normal forms (GNF). 1. $S \rightarrow AB, A \rightarrow aA \mid bB \mid b, B \rightarrow b$. 2. $S \rightarrow CA \mid BB, B \rightarrow b \mid SB, C \rightarrow b, A \rightarrow a$.	10	3	3
Q3 A	Construct the Context free grammar for the following with justification. 1. String starting and ending with different letter over the alphabet (a, b). 2. To generate the language $L = \{a^n b^{n+1} \mid n \geq 1\}$ 3. To generate the palindrome over the alphabet (a, b) 4. To generate the language $L = \{a^m b^{m-3} \mid m \geq 3\}$	10	3	3

B	For each of the following production rules, decide whether they follow the format of Chomsky Normal Form or not. If not, give an equivalent set of production rules which follow the format of Chomsky Normal Form. (1)S->AS (2)S->AAS (3)A->SA (4)A->aa	10	3	3
Q4	State True or False with appropriate justification (i) Context free grammars and push down automata are equivalent in power. They recognise the same class of languages. (ii) All regular languages are context-free. (iii) The stack of a PDA is with last-in-first-out order. (iv) The class of context-free languages is closed under union.	20	4	3
Q5A	State True or False with appropriate justification (i) Every Turing Machine always halts. (ii) If a language A is recognized by a Turing Machine M and the complement of A is recognized by a Turing Machine N, A is decidable. (iii) A Turing Machine may halt even before it reaches the end of an input string. (iv) If a Turing Machine loops indefinitely on a string w, it will loop indefinitely on all strings.	10	4	2
B	(a) Define recursive and recursively enumerable languages. Give an example of each. (b) Give an example of a language which is recursively enumerable but not recursive. (c) Is every recursive language also recursively enumerable. Explain why / why not.	10	4	3